

A Data Fusion Framework for Covariate Prioritization in Incomplete Short-Term Soil Nitrogen Records

Conditional effects, importance-method disagreement, and predictive limits in a five-year grazing and cover crop trial

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INTRODUCTION

Long-term trials are the benchmark for evaluating grazing and cover crop systems, but most soil records are short and incomplete, with measurements available only for certain years, treatments, or sampling times.

Remote sensing and machine learning now make it possible to integrate such records and use them for prediction. Whether a short-term analysis, i.e., a five-year record, provides sufficient information for reliable prediction is rarely evaluated.

OBJECTIVES

1 Audit

Which variables have sufficient temporal and treatment coverage?

2 Rank

Which fused covariates explain soil N, and do rankings method agree?

3 Test

Can these covariates predict an unseen year?

METHODS

Study area

9 one-acre plots · 3 randomized treatments (unfertilized triticale, fertilized triticale, triticale + legume) · grazed and ungrazed plots · 0–10 and 10–20 cm · 2019–2023 · 1,363 soil samples

Coverage audit

	2019	2020	2021	2022	2023	Role
NH ₄ ⁺ / NO ₃ ⁻	Complete	Complete	Complete	Complete	Complete	Response
GHG flux	All treatments	All treatments	All treatments	All treatments	All treatments	Covariate
pH / EC / BD	All treatments	Plot IDs lost	All treatments	Time 2 only	Time 2 only	Baseline
Carbon fractions	All treatments	All treatments	All treatments	Time 2 only	Time 2 only	Snapshot
FAME	All treatments	All treatments	All treatments	Time 2 only	Time 2 only	Snapshot
■ Complete ■ All treatments ■ Time 2 only ■ Plot IDs lost ■ Not collected						

Analysis

Covariates — NDVI, EVI and SAR-VV via Google Earth Engine; SPEI-90d per sampling window (GRIDMET); chamber CO₂, N₂O and CH₄ as seasonal means.

Ranking — RF impurity (training splits; favors continuous over binary predictors), permutation (held-out MSE; model-agnostic) and GPR ARD length-scales. Top five carried forward.

Grazing effect — Mann–Whitney U within treatment × year; Holm-corrected within the 2023 endpoint family.

Model — One global GPR, n = 264; log-transformed target; Matérn ARD kernel; treatment, grazing and depth as inputs, not filters.

Validation — Fit 2019–2022, predict 2023; RMSE, R², and 95% interval coverage reported with width.

FIVE CHECKS BEFORE BUILDING A FORECASTING PIPELINE ON A SHORT RECORD

1 Let coverage decide each variable’s role, before modelling anything

Two years at one time point support a comparison, not a trajectory. High-quality data can only do so much if it isn’t collected consistently.

2 Test conditional variables directly before ranking discards them

Grazing ranked last or near-last under every method we tried, yet held our only significant management effect.

3 Rank covariates at least two ways, and report the disagreement

CO₂ flux ranked third in-sample while *degrading* held-out prediction — visible only under permutation.

4 Match covariate resolution to your sampling design

An annual SPEI index on multi-season sampling is collinear with year by construction. Extracted per sampling window, ours became testable.

5 Run the temporal holdout first — it decides whether the rest is worth building

Report interval coverage *with* width. Ours reached 100% coverage only because the intervals exceeded the full observed range.

DISCUSSION

• A covariate can carry a real, conditional effect without being a useful global predictor.

• Importance ranking demonstrates this: grazing ranked last or near-last under all three methods. However, it is an important covariate within the different management systems when analyzed separately.

• To address the issue of limited soil data, grouping folds by plot ID — as Lucero et al. (2026) did — may suit sparse soil records better than the year-grouped folds used here, though it tests generalization to new plots rather than to new years.

• Due to the limitation of the soil record, five annual campaigns can only support description and association, not prediction.

• Another limitation is the small plot size relative to the drought index grid. Nine one-acre plots share a single 4.6 km pixel, so SPEI varies in time but not in space, unlike the 10 m vegetation and SAR covariates.

RESULTS

Year and water balance rank highest; grazing ranks last

RF and ARD are shares of total relevance · Perm is change in held-out MSE, where negative means shuffling improved prediction

	NH ₄ ⁺ · RF	NH ₄ ⁺ · Perm	NH ₄ ⁺ · ARD	NO ₃ ⁻ · RF	NO ₃ ⁻ · Perm	NO ₃ ⁻ · ARD
Water balance	0.159	0.00	0.191	0.174	0.00	0.198
Year	0.239	0.00	0.141	0.133	0.00	0.152
NDVI	0.138	+1.38	0.547	0.104	+0.34	0.320
CO ₂ flux	0.155	−6.98	0.012	0.109	−2.48	0.149
Depth	0.019	+3.30	0.021	0.108	+37.63	0.040
Treatment	0.020	+2.87	0.054	0.027	+0.16	0.061
Grazing	0.004	−0.32	0.012	0.008	−0.18	0.035

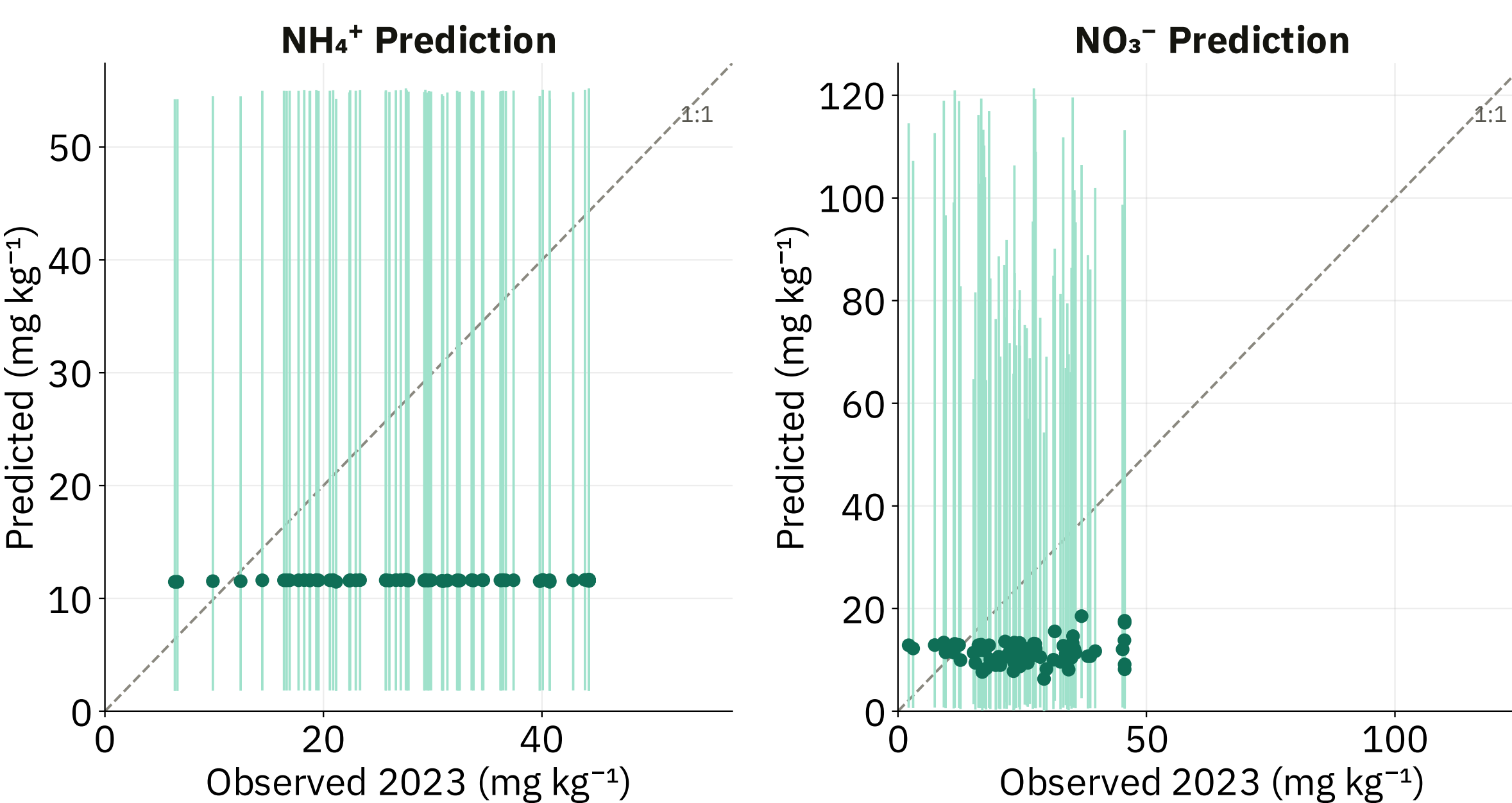
Yet grazing depleted N where nothing replaced it

Grazed minus ungrazed, mg kg^{−1} · Mann–Whitney U

	2020	2021	2022	2023
Unfertilized · NH₄⁺	+1.1	−0.2	−2.6	−11.5**
Unfertilized · NO₃⁻	+4.0	+2.9*	−2.9	−15.0**
Fertilized · NH ₄ ⁺	−1.9*	−2.9	+1.1	+5.3
Fertilized · NO ₃ ⁻	−2.5	+4.7*	−1.5	+7.5
Legume · NH ₄ ⁺	—	+0.1	−4.1	+4.7
Legume · NO ₃ ⁻	—	+1.8	−9.9	+3.3

** Holm-corrected p < 0.05 (2023, 6 tests) · * uncorrected p < 0.05 · blue = grazing lowers N · legume plots not sampled in 2020

And the model could not predict a held-out year



NH₄⁺ · 2023 holdout
R² −3.26 · 53 mg kg^{−1} wide

NO₃⁻ · 2023 holdout
R² −1.83 · 88 mg kg^{−1} wide

NEXT STEPS

Extend the temporal record · add plot-scale moisture · nest feature selection within training folds · close temporal sampling gaps · let covariates carry the time signal



View Tableau Dashboard
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